Worksheet: the Gushchin construction for the grid solver

Companion to `python/shared/mixture_sb.py`, `paper_route_sb.py` and `docs/sb_primer.tex`, Sections 9–10

24 October 2026

Goal for today. Be able to (i) state in one sentence why a constructed pair has a known Schrödinger bridge, (ii) derive every formula in the two reference classes by hand, (iii) audit each line of the code against your derivation, and (iv) run the 2D split example and the ring example through the GPU solver and know what error to expect. Parts A–E are pen, paper and plain numpy (no JAX, runs on the Mac). Part F is the GPU plan. Answers are in Appendix A; try each exercise before looking.

Conventions

Solver PDE: $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$, $m = \rho v$, on $[0,1]^d$ with no-flux walls, $t \in [0,1]$. Reference SDE $dX = \sqrt{2\varepsilon}\, dW$, so the heat kernel over time t is $\mathcal{N}(\cdot;\, x,\, 2\varepsilon t\, I)$. We write $\sigma^2 := 2\varepsilon$. The paper's regularization ϵ_{p} equals $\sigma^2 = 2\varepsilon$: *their* ε is twice *ours*. The code writes `2*eps` wherever σ^2 appears. Primer section numbers refer to `sb_primer.pdf`.

Part A. The one fact everything rests on

Exercise 1 *The Gibbs form (primer Prop. 5.4 and Prop. 9.1, paper Thm. 3.2).*

Write down the plan

$$\pi(x,y) = \hat{\varphi}(x)\, \mathcal{K}_\sigma(x,y)\, \varphi(y), \qquad \mathcal{K}_\sigma(x,y) = \mathcal{N}(y; x, \sigma^2 I),$$

for two positive functions $\hat{\varphi}, \varphi$ with $\iint \pi = 1$. Compute its two marginals in terms of $\hat{\varphi}$, φ and the heat flow, and name the two functions

$$\varphi_0(x) := \int \mathcal{K}_\sigma(x,y)\varphi(y)\, dy, \qquad \hat{\varphi}_1(y) := \int \hat{\varphi}(x)\mathcal{K}_\sigma(x,y)\, dx.$$

State in one sentence why π is the entropic-optimal plan between its own marginals. (Hint: the optimal plan is characterized by having the form "function of x times kernel times function of y"; nothing else is needed.)

Exercise 2 *From the static plan to the bridge.*

Define the backward and forward heat flows of the two factors,

$$\varphi_t := \mathcal{K}_{\sigma\sqrt{1-t}} * \varphi \;\; (\partial_t \varphi_t = -\varepsilon \Delta \varphi_t,\; \varphi_1 = \varphi), \qquad \hat{\varphi}_t := \mathcal{K}_{\sigma\sqrt{t}} * \hat{\varphi} \;\; (\partial_t \hat{\varphi}_t = +\varepsilon \Delta \hat{\varphi}_t,\; \hat{\varphi}_0 = \hat{\varphi}),$$

and set $\rho_t := \varphi_t \hat{\varphi}_t$, $v := \sigma^2 \nabla \log \varphi_t = 2\varepsilon \nabla \varphi_t / \varphi_t$. Show by direct differentiation that

$$\partial_t \rho_t + \nabla \cdot (\rho_t v) = \varepsilon \Delta \rho_t.$$

This is the computation that ties the construction to *our* PDE; do it fully. (Expand $\Delta(\varphi\hat{\varphi}) = \hat{\varphi}\Delta\varphi + 2\nabla\varphi\cdot\nabla\hat{\varphi} + \varphi\Delta\hat{\varphi}$ and $\nabla \cdot (\rho v) = 2\varepsilon\, \nabla \cdot (\hat{\varphi}\nabla\varphi)$.) Then check $\rho_0 = \hat{\varphi}\varphi_0$ and $\rho_1 = \varphi\hat{\varphi}_1$ match Exercise 1.

Exercise 3 *Which object is which.*

Fill in the table from memory, then check against the primer's symbol table (Sec. 11).

object	paper's name/symbol	code
φ (the y-side factor)	$\exp(f^*/\epsilon_{\mathrm{p}})$, LSE potential, eq. (15)	`phi(1.0, .)`, args `(w, b, Q)`
φ_0	_____	_____
$\hat{\varphi}$	_____	_____
$\pi^*(y\|x)$	_____	_____
$v = \sigma^2 \nabla \log \varphi_t$	_____	_____

Part B. The two Gaussian identities

Everything explicit below comes from two facts about $\mathcal{N}(z; \mu, \Sigma)$.

Exercise 4 *Convolution rule.*

Show $\mathcal{N}(\cdot; \mu, \Sigma) * \mathcal{N}(\cdot; 0, \tau^2 I) = \mathcal{N}(\cdot; \mu, \Sigma + \tau^2 I)$ (sum of independent Gaussians). Conclude: the heat flow with generator $\varepsilon\Delta$ over a time s adds $2\varepsilon s\, I$ to every covariance. Which sign of s goes with φ_t and which with $\hat{\varphi}_t$? Write both φ_t and $\hat{\varphi}_t$ for mixtures $\varphi = \sum_n w_n \mathcal{N}(\cdot; b_n, Q_n)$, $\hat{\varphi} = \sum_i u_i \mathcal{N}(\cdot; a_i, P_i)$.

Code check: `mixture_sb.py, phi / phihat`

Check `Qn + 2*self.eps*(1-t)*I` in `phi` and `Pi + 2*self.eps*t*I` in `phihat` against your answer. Which one is wrong if you accidentally use the paper's ϵ_{p} in place of ε? (Both. By a factor 2 in the inflation.)

Exercise 5 *Product rule.*

Show that for the same variable z,

$$\mathcal{N}(z; a, A)\, \mathcal{N}(z; b, B) = \mathcal{N}(a; b, A+B)\, \mathcal{N}\big(z;\, c,\, C\big), \quad C = (A^{-1} + B^{-1})^{-1}, \quad c = C(A^{-1}a + B^{-1}b).$$

Derive c and C by completing the square in z; derive the constant $\mathcal{N}(a; b, A+B)$ by integrating both sides over z (the left side's integral is the convolution rule evaluated at $a - b$). Interpret c as a precision-weighted average.

Code check: `mixture_sb.py, components`

Lines `Pt = inv(Pi + 2 eps t I)`, `Qt = inv(Qn + 2 eps (1-t) I)`, `C = inv(Pt + Qt)`, `c = C @ (Pt @ ai + Qt @ bn)`. Match each to A^{-1}, B^{-1}, C, c. Note that `Pt`, `Qt` are already *inverses*; the names are a small trap.

Part C. Recipe A: the paper route on a grid (`paper_route_sb.py`)

Choices: ρ_0 (any smooth positive density), $\varphi = \sum_n w_n \mathcal{N}(\cdot; b_n, Q_n)$. Forced: everything else.

Exercise 6 *Why $\hat{\varphi}$ is forced.*

Using Exercise 1, show that the x-marginal of $\hat{\varphi}\mathcal{K}_\sigma\varphi$ equals a prescribed ρ_0 if and only if $\hat{\varphi} = \rho_0/\varphi_0$. Explain why the normalizing constant is then $K = 1$ automatically (what is $\iint \pi$?). Then explain why, for the ring ρ_0, $\hat{\varphi}$ is a *crescent*: where is φ_0 smallest on the ring? The ring is not a sharp annulus but a smooth Gaussian shell (`ring_rho0`),

$$\rho_0(x) \propto \exp\Big(-\frac{(|x-c|-r)^2}{2w^2}\Big), \qquad c = (0.35, 0.50),\ r = 0.12,\ w = 0.025,$$

peaked on the circle $|x - c| = r$ and falling off over a width w on either side; it is smooth and strictly positive, which Recipe A's division and heat flow and the solver's $O(h^2)$ accuracy both need. Along the centreline circle it is constant, which is all the argument uses.

Code check: `paper_route_sb.py, __init__`

`self.phihat0_fine = self.rho0_fine / self.phi(0.0, XF)`. Confirm: `phi(0.0,.)` is φ_0 (backward flow all the way to $t = 0$, covariance $Q_n + 2\varepsilon I$), and `rho0_fine` is normalized on the *fine* grid, which is what makes $K = 1$.

Exercise 7 *The one numerical step.*

$\hat{\varphi}_t = \mathcal{N}(\cdot; 0, 2\varepsilon t I) * \hat{\varphi}$ is computed by midpoint quadrature on an $n_f \times n_f$ grid. Show that for a tensor grid of evaluation points $\{x_p\} \times \{y_q\}$ the convolution of a function F known on the fine grid is $K_x F K_y^\top$ with

$$(K_x)_{pj} = \frac{h_f}{\sqrt{2\pi \cdot 2\varepsilon t}} \exp\Big(-\frac{(x_p - x_j^f)^2}{2 \cdot 2\varepsilon t}\Big).$$

Why does the Gaussian kernel make this separable? Why does the midpoint rule converge spectrally here, and what is the smoothness scale of the integrand at the earliest solver time $t = 1/n_t$?

Code check: `paper_route_sb.py, phihat / kmat`

`var = 2*self.eps*t`; `kmat` returns `exp(-d*d/(2*var))/sqrt(2*pi*var) * self.hf`. Match each factor to $(K_x)_{pj}$. Check that the special case `t == 0.0` returns $\hat{\varphi}$ itself (the kernel would be a delta). The normalization of the discrete kernel is *not* enforced (rows do not sum to exactly 1); quantify this in Exercise 8.

Exercise 8 *Numerical check of the quadrature.*

Run this on the Mac. It builds the ring example at two fine resolutions and compares $\hat{\varphi}_t$ at the solver's cell centres at the earliest time. You should see a difference below 10^{-9} relative to the peak once $n_f \geq 256$.

```
import sys, numpy as np; sys.path.insert(0, 'python')
from shared.paper_route_sb import example_ring_to_two_2d
n, nt = 64, 32
xc = (np.arange(n) + 0.5) / n
a = example_ring_to_two_2d(eps=0.01, nfine=256).phihat(1/nt, xc, xc)
```

```
b = example_ring_to_two_2d(eps=0.01, nfine=512).phihat(1/nt, xc, xc)
print(np.abs(a - b).max() / b.max())
```

Also print `a.sum()*(1/n)**2` against `phihat(0, xc, xc).sum()*(1/n)**2`: the heat flow preserves the integral of $\hat{\varphi}$. Do they agree? To what accuracy, and why not exactly (two reasons: box tails, and the discrete kernel rows)?

Exercise 9 *The paper's Proposition 3.3 is the conditional plan.*

From $\pi^*(y|x) = \varphi(y)\mathcal{K}_\sigma(x,y)/\varphi_0(x)$ and the product rule, derive that $\pi^*(\cdot|x)$ is a Gaussian mixture in y with

$$\text{cov } S_n = (Q_n^{-1}+\sigma^{-2}I)^{-1}, \quad \text{mean } m_n(x) = S_n(Q_n^{-1}b_n+\sigma^{-2}x), \quad \text{weight } \propto w_n\mathcal{N}(x; b_n, Q_n+\sigma^2 I).$$

Now translate to the paper: with $Q_n^{-1} = A_n/\epsilon_\mathrm{p}$ and $\sigma^2 = \epsilon_\mathrm{p}$, show $S_n = \epsilon_\mathrm{p}(A_n + I)^{-1}$ and $m_n(x) = (A_n + I)^{-1}(A_n b_n + x)$, their Σ_n and $\mu_n(x)$. What does "sampling from the plan" mean in these terms, and why does the paper never need the density of ρ_1?

Code check: `mixture_sb.py, conditional_mixture`

`s2 = 2*self.eps; Cn = inv(inv(Qn) + I/s2); mn = Cn @ (solve(Qn, bn) + x/s2); ws.append(wn * gauss(x, bn, Qn + s2*I))`. Match to S_n, $m_n(x)$, and the weight. This method is shared by both recipes because it depends on φ only.

Exercise 10 *Why Recipe A cannot give ρ_1 in closed form.*

$\rho_1 = \varphi\,\hat{\varphi}_1$ with $\hat{\varphi}_1 = \mathcal{K}_\sigma * (\rho_0/\varphi_0)$. Even when ρ_0 is one Gaussian and φ a two-component mixture, explain in one sentence what blocks a closed form. (Primer Example 9.2 does the 1D case.)

Part D. Recipe B: both factors as mixtures (`mixture_sb.py`)

Choices: $\hat{\varphi} = \sum_i u_i\mathcal{N}(\cdot; a_i, P_i)$ and $\varphi = \sum_n w_n\mathcal{N}(\cdot; b_n, Q_n)$. Forced: *both* marginals.

Exercise 11 *The switch, and why it is legitimate.*

Recipe A fixes (ρ_0, φ); Recipe B fixes $(\hat{\varphi}, \varphi)$. Which part of Exercise 1 says this is allowed? What do we lose (which object is no longer a free choice)? What do we gain (which object becomes explicit)?

Exercise 12 *Branch masses.*

Write $\pi = \hat{\varphi}\mathcal{K}_\sigma\varphi/K$ as a double sum over (i,n) and show that the mass of term (i,n) is

$$\Omega_{in} = \frac{u_i w_n\,\mathcal{N}(a_i; b_n, P_i + Q_n + \sigma^2 I)}{K}, \qquad K = \sum_{i,n}(\text{numerators}).$$

(Integrate over y first with the convolution rule, then over x with the product rule's constant.) Why is Ω_{in} independent of t? Why is it exactly "the mass that travels from source blob i to target blob n"?

Code check: `mixture_sb.py, __init__`

`num[i,n] = ui*wn*gauss(ai, bn, Pi + Qn + 2*eps*I);` `self.K = num.sum();` `self.omega = num/K`. Match to Ω_{in}. Then run:

```
from shared.mixture_sb import example_split_3d
print(np.round(example_split_3d().omega, 3))
```

and reproduce the 2×3 table in `candidate_3d_problems.tex`: 0.391/0.006/0.103 per row. Say in words what the 0.006 entry is.

Exercise 13 *The bridge density.*

Apply the product rule to one term $u_i \mathcal{N}(z; a_i, P_i + \sigma^2 t I) \cdot w_n \mathcal{N}(z; b_n, Q_n + \sigma^2(1-t)I)$ and show

$$\rho_t(z) = \sum_{i,n} \Omega_{in}\, \mathcal{N}\big(z; c_{in}(t), C_{in}(t)\big), \quad C_{in}(t) = \big[(P_i + \sigma^2 t I)^{-1} + (Q_n + \sigma^2(1-t)I)^{-1}\big]^{-1},$$

$c_{in}(t) = C_{in}(t)\big[(P_i + \sigma^2 t I)^{-1} a_i + (Q_n + \sigma^2(1-t)I)^{-1} b_n\big]$. Key step: the product-rule constant is $\mathcal{N}(a_i; b_n, P_i + Q_n + \sigma^2 I)$ for every t. Why?

Exercise 14 *Numerical identity checks (run on the Mac).*

```
from shared.mixture_sb import example_split_2d
sb = example_split_2d(eps=0.01)
z = np.random.default_rng(0).uniform(0.1, 0.9, size=(2000, 2))
for t in (0.0, 0.3, 1.0):
    lhs = sb.rho(t, z)
    rhs = sb.phi(t, z) * sb.phihat(t, z) / sb.K
    print(t, np.abs(lhs - rhs).max() / lhs.max())
```

Expect 10^{-15}. Then check the Fokker–Planck residual yourself by finite differences on a 400^2 grid at $t = 0.5$ using `sb.rho` and `sb.momentum`; expect $O(h^2)$ (the doc records $2.3 \cdot 10^{-3}$ at 200^2 falling by 4 per refinement).

Exercise 15 *The drift is not a branch average.*

Write $v = 2\varepsilon \nabla \log \varphi_t$ for the mixture $\varphi_t = \sum_n w_n \mathcal{N}(\cdot; b_n, S_n(t))$, $S_n(t) = Q_n + \sigma^2(1-t)I$, as a responsibility-weighted sum of per-component pulls $-S_n(t)^{-1}(z - b_n)$. Explain why v depends on φ only, not on $\hat{\varphi}$, and why it is *not* $\sum_{i,n} \Omega_{in}\, v_{in}$ with v_{in} the single-Gaussian bridge velocities. (Which object has the i index at all?)

Code check: `mixture_sb.py, velocity`

`g = wn*gauss(z, bn, S);` `num += g * (-solve(S, z - bn));` `den += g;` `return 2*eps*num/den`. Match to your formula. Check the sign: at a point z to the left of a lone target b_n the drift must point right.

Exercise 16 *Reduction to the Gaussian closed form.*

With $I = N = 1$, $P = Q = s^2 I$, $a \neq b$: show ρ_t is one Gaussian with covariance $C(t) = \dfrac{(s^2 + \sigma^2 t)(s^2 + \sigma^2(1-t))}{2s^2 + \sigma^2}\, I$ and centre moving linearly from $c(0)$ to $c(1)$. The solver's reference

`analytical_sb_gaussian_2d` writes the variance as $\sigma_g^2 + 2\alpha t(1-t)$ with $\alpha = \sqrt{\sigma_g^4 + \varepsilon^2} - \sigma_g^2$, where $\sigma_g^2 = C(0)$ is the *marginal* variance. Show the two expressions agree (expand $C(t)$ as a quadratic in t and identify the coefficient of $t(1-t)$ as 2α after eliminating s^2 in favour of σ_g^2). This is why "$I = N = 1$ reproduces Bunne" is a real check and not a tautology.

Exercise 17 *Reading a design.*

For `example_split_2d`: $\hat{\varphi} = \mathcal{N}((0.30, 0.50), 0.05^2 I)$, two targets at $(0.70, 0.30)$ and $(0.70, 0.70)$ with $Q = 0.04^2 I$, $\varepsilon = 0.01$. Without running code, predict: (a) is ρ_0 centred at 0.30? (b) is its width 0.05? (c) are the two Ω's equal? Then run `sb.components(0.0)` and compare with the doc's table (centre 0.342, std 0.047). Explain each shift via the precision-weighted average c.

Part E. Code audit checklist

Go through these with the files open. Tick each or write down what you found.

1. `gauss`: Cholesky-based; returns a *normalized* density with $(2\pi)^{-D/2} \det^{-1/2}$. Confirm the paper's $Q(\cdot|b, A)$ bells are unnormalized and that this is why the weight dictionary carries det factors (primer Sec. 9, dictionary item 2). Which code path would break if `gauss` were unnormalized? (`omega`: the branch masses would be wrong, though `rho` would still be proportional to the right thing.)

2. Every covariance inflation is `2*eps*(...)`: `phi`, `phihat`, `components`, `velocity`, `conditional_plan`, `conditional_mixture`, and in `paper_route_sb.py` `phi`, `grad_log_phi`, `phihat`, `conditional_plan`. Grep for a bare `eps` used as a variance.

3. `staggered_reference`: ρ at times $k\,dt$, $k = 1, \ldots, n_t - 1$ (interior time edges, where the solver's ρ lives); m at $(k - \frac{1}{2})dt$, $k = 1, \ldots, n_t$ (time cell centres); m_d on interior d-faces $x = j\,h$, $j = 1, \ldots, n_d - 1$, cell centres in the other directions. Compare with the layout stated in the docstring of `analytical_sb_gaussian_2d`: `rho (ntm,nx,ny)`, `mx (nt,nxm,ny)`, `my (nt,nx,nym)`.

4. `staggered_reference` renormalizes ρ per time slice on the grid before forming $m = \rho v$ (both classes). Is this consistent with how `setup_problem_2d` treats ρ_0, ρ_1 (`rho0 = rho0_pdf/(sum*dx*dy)`)? What is the size of the correction (tail mass, $\sim 10^{-7}$)? For m the renormalization is done on the *face* grid of each component separately; argue it is harmless at that level.

5. `pdf_func(t)` returns a callable on broadcast coordinate arrays, which is exactly what `ProblemDef2D.rho0_fun yy[None,:])` passes in. It converts with `np.asarray`, so JAX arrays from the setup are accepted. Check the `ProblemDef2D` fields `mu0`, `mu1`, `sigma` are only used by `analytical_sb_gaussian_2d` and nothing in the solver path (grep `2d/setup`).

6. `FixedRho0SB` is 2D-only (`self.D = 2`, tensor `meshgrid` of two axes). `MixtureSB` is dimension-agnostic. Note this before planning a 3D ring run.

7. Box tails. Both classes assume $\mathbb{R}^d$. For each example, compute the mass of ρ_t outside $[0, 1]^d$ at $t = 0, \frac{1}{2}, 1$ (the doc quotes $\leq 2 \cdot 10^{-7}$ in 2D). This bounds the Neumann-vs-free-space mismatch of the reference.

Part F. Running it on the GPU today

Nothing has been run through the solver yet. The plan below mirrors `2d/scripts/sb_gaussian_2d.py`, which is the right template.

1. **Problem definition.** Build the solver's problem from the reference:

```
from shared.mixture_sb import example_split_2d
sb = example_split_2d(eps=vareps)
prob_def = ProblemDef2D(name="mixsplit", rho0_func=sb.pdf_func(0.0),
                        rho1_func=sb.pdf_func(1.0), mu0=(0,0), mu1=(0,0), sigma
    =0.0)
problem = setup_problem_2d(prob_def, nt=n, nx=n, ny=n)
```

   The same ε must go into the reference and into the projection; they are the same physical constant here, not a regularizer chosen afterwards.

2. **Solver settings.** Use the jitted path with the DCT per-mode projection (`precomp_expsemi_proj_2d(problem vareps, method="dct")`) as in `run_suite_2d.py`. Take $\gamma = 0.1$, not 1: the 2D density peak is of order 60 here too and $\gamma = 1$ was found to be about $8\times$ too stiff. Budget 20,000 iterations at $n \le 128$, as for the Gaussian suite.

3. **Reference on the staggered grid.** `rho_ref, mx_ref, my_ref = sb.staggered_reference(nt=n, ns=(n, n))`. Compare with `result.rho_stag`, `mx_stag`, `my_stag` in the relative Frobenius norm, exactly as `sb_gaussian_2d.py` does.

4. **What to expect.** The error should fall as $O(h^2)$ over $n = 32, 48, 64, 96, 128$ (the Gaussian suite gave orders near 2 for ρ and m at $\varepsilon = 10^{-2}$). A *new* diagnostic available here only: integrate the solver's ρ_1 and ρ_t over the half-planes $y < 0.5$, $y > 0.5$ and compare with the branch masses $\Omega_{1,1} = \Omega_{1,2} = 0.5$ at every t. Mass transfer between branches is a qualitative failure mode the Gaussian test cannot see.

5. **Paper-route run.** Repeat with `example_ring_to_two_2d(eps, nfine=512)` and `sb.staggered_reference( nx=n, ny=n)` (note the different signature). Here ρ_0 is a ring and $\hat{\varphi}$ a crescent; the solver has to move the far side of the ring the long way round. Expect it to need more iterations than the split example.

6. **A ρ_0 with no Gaussian in it.** Recipe A does not care what ρ_0 is. `example_cross_to_two_2d` starts from a plus-sign: two smoothed rectangles, each a product of error-function differences (`smoothed_box`), so it has a flat top, straight edges rounded over $w = 0.012$, and four concave corners; it is closed form but no mixture of anything. Checks on the Mac: mass $1 - 4 \cdot 10^{-7}$ on the box, quadrature converged at $n_f = 256$ to 10^{-15}, finite-difference Fokker–Planck residual $1.9 \cdot 10^{-3}, 4.9 \cdot 10^{-4}, 1.2 \cdot 10^{-4}$ on $200^2, 400^2, 800^2$, i.e. $O(h^2)$; $\hat{\varphi}$ varies by $\sim 10^5$ across the support, more than the ring, so the far arm of the cross carries most of the prefactor (Fig. 1). Same run protocol as the ring; the corners and the plateau are what a Gaussian test never exercises.

7. **3D.** `example_split_3d(eps=0.005)` through `3d/` at $n = 32, 48, 64$, reference from `staggered_reference(nt= ns=(n,n,n))`. The 1% cross branch $\Omega_{1,2} = 0.006$ is the stress test: check the solver's mass in the region between source blob 1 and target b_2 at $t = 0.75$ against the reference's.

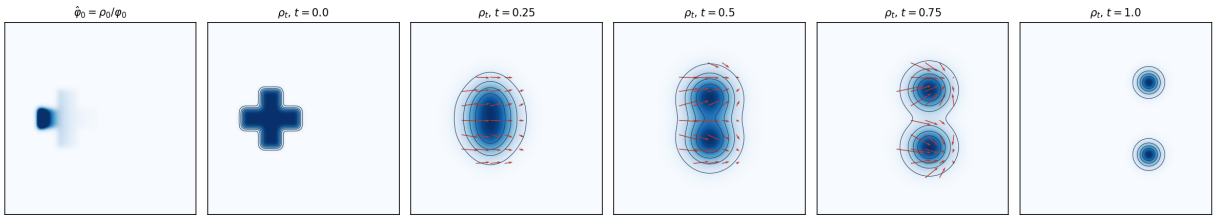


Figure 1: Plus-sign to two blobs, $\varepsilon = 0.01$, all from the Recipe A reference (no solver run). Left: the forced factor $\hat{\varphi}$, concentrated on the arm facing away from the targets. Then ρ_t at $t = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ with the closed-form drift as arrows; contours at 10–90% of each panel's peak.

8. **Write-up.** One table (grid, ρ error, m error, order, iterations, time) per example, the branch-mass check as a short second table, and one figure of ρ_t slices with the reference contours overlaid. State the two reference error sources (quadrature for Recipe A, box tails for both) with their sizes, so the reader knows the reference is $10^4\times$ more accurate than the solver.

A Answers and hints

Ex. 1. $\int \pi\, dy = \hat{\varphi}(x)\varphi_0(x)$ and $\int \pi\, dx = \hat{\varphi}_1(y)\varphi(y)$. Optimality: the entropic plan between two marginals is the unique coupling of the form $f(x)\mathcal{K}_\sigma(x,y)g(y)$ (Schrödinger system / Sinkhorn fixed point), so any such product *is* the optimal plan for the marginals it happens to have.

Ex. 2. $\partial_t \rho = \hat{\varphi}\partial_t\varphi + \varphi\partial_t\hat{\varphi} = -\varepsilon\hat{\varphi}\Delta\varphi + \varepsilon\varphi\Delta\hat{\varphi}$. And $\nabla\cdot(\rho v) = 2\varepsilon\nabla\cdot(\hat{\varphi}\nabla\varphi) = 2\varepsilon(\nabla\hat{\varphi}\cdot\nabla\varphi + \hat{\varphi}\Delta\varphi)$. Sum: $\varepsilon(\hat{\varphi}\Delta\varphi + 2\nabla\hat{\varphi}\cdot\nabla\varphi + \varphi\Delta\hat{\varphi}) = \varepsilon\Delta(\varphi\hat{\varphi})$.

Ex. 3. φ_0: paper's normalizer Z_x (up to a constant), code `phi(0.0,.)`. $\hat{\varphi}$: never written in the paper; code `phihat(0,.)` (Recipe B) or `phihat0` (Recipe A). $\pi^*(y|x)$: paper eq. (14) and Prop. 3.3; code `conditional_plan`, `conditional_mixture`. v: paper Cor. 3.5 eq. (16); code `velocity`.

Ex. 4. *Convolution rule.* Let $X \sim \mathcal{N}(\mu, \Sigma)$ and $Y \sim \mathcal{N}(0, \tau^2 I)$ be independent. The density of $X + Y$ is the convolution of the two densities. But $X + Y$ is a linear combination of jointly Gaussian variables, hence Gaussian, with

$$\mathbb{E}[X+Y] = \mu, \qquad \operatorname{Cov}(X+Y) = \operatorname{Cov} X + \operatorname{Cov} Y + 2\operatorname{Cov}(X,Y) = \Sigma + \tau^2 I,$$

the cross term vanishing by independence. A Gaussian is determined by mean and covariance, so $\mathcal{N}(\cdot;\mu,\Sigma) * \mathcal{N}(\cdot;0,\tau^2 I) = \mathcal{N}(\cdot;\mu,\Sigma+\tau^2 I)$. The same argument with $Y \sim \mathcal{N}(\mu_2,\Sigma_2)$ gives the general rule $\mathcal{N}(\cdot;\mu_1,\Sigma_1) * \mathcal{N}(\cdot;\mu_2,\Sigma_2) = \mathcal{N}(\cdot;\mu_1+\mu_2,\Sigma_1+\Sigma_2)$: means add and covariances add. Only the $\mu_2 = 0$ case is needed below, because the heat kernel has mean zero: $\mathcal{K}_\sigma(x,y) = \mathcal{N}(y;x,\sigma^2 I)$ looks centred at x, but x is the convolution variable, and $\mathcal{N}(y;x,\sigma^2 I) = \mathcal{N}(x - y;0,\sigma^2 I)$, so $\int \mathcal{K}_\sigma(x,y)\varphi(y)\,dy = (\mathcal{N}(\cdot;0,\sigma^2 I) * \varphi)(x)$. Probabilistically the increment $\sqrt{2\varepsilon}\,W_s$ has mean zero; a nonzero mean would appear only if the reference SDE had a drift, which would translate every component centre. (Fourier proof, no probability: the characteristic function of $\mathcal{N}(\mu,\Sigma)$ is $\exp(ik^\top\mu - \frac{1}{2}k^\top\Sigma k)$; convolution multiplies transforms, so the exponents $-\frac{1}{2}k^\top\Sigma k$ and $-\frac{1}{2}\tau^2|k|^2$ add.)

Heat flow. The solution of $\partial_s u = \varepsilon \Delta u$, $u(0) = u_0$, is $u(s) = G_s * u_0$ with $G_s(z) = (4\pi\varepsilon s)^{-D/2} e^{-|z|^2/4\varepsilon s}$. Since $4\pi\varepsilon s = 2\pi \cdot (2\varepsilon s)$ and $|z|^2/4\varepsilon s = |z|^2/(2 \cdot 2\varepsilon s)$, this is $G_s = \mathcal{N}(\cdot; 0, 2\varepsilon s I)$, the density of $\sqrt{2\varepsilon}\, W_s$ for the reference SDE $dX = \sqrt{2\varepsilon}\, dW$. By the rule with $\tau^2 = 2\varepsilon s$, a Gaussian $\mathcal{N}(\mu, \Sigma)$ flows to $\mathcal{N}(\mu, \Sigma + 2\varepsilon s I)$, and by linearity a mixture flows term by term with unchanged weights.

Signs. $\hat{\varphi}$ runs forward, $\partial_t \hat{\varphi}_t = +\varepsilon \Delta \hat{\varphi}_t$, $\hat{\varphi}_0 = \hat{\varphi}$: elapsed time $s = t$, so $\hat{\varphi}_t = \mathcal{N}(0, 2\varepsilon t I) * \hat{\varphi}$. φ runs backward, $\partial_t \varphi_t = -\varepsilon \Delta \varphi_t$, $\varphi_1 = \varphi$: with $s = 1 - t$ this is a forward heat equation started from φ at $s = 0$, so $\varphi_t = \mathcal{N}(0, 2\varepsilon(1-t) I) * \varphi$. Each factor is sharpest at the end where it is prescribed and blurrier away from it.

Mixtures.

$$\varphi_t(z) = \sum_n w_n \mathcal{N}\big(z; b_n, Q_n + 2\varepsilon(1-t) I\big), \qquad \hat{\varphi}_t(z) = \sum_i u_i \mathcal{N}\big(z; a_i, P_i + 2\varepsilon t I\big).$$

Two values recur later: φ_0 (covariances $Q_n + 2\varepsilon I$) is the paper's normalizer Z_x; $\hat{\varphi}_1$ (covariances $P_i + 2\varepsilon I$) is what multiplies φ to give ρ_1. Using the paper's $\epsilon_{\mathrm{p}} = 2\varepsilon$ in place of ε would double both inflations; the symptom is a reference that fails the finite-difference Fokker–Planck check against our PDE with coefficient ε.

Ex. 5. Exponent in z: $-\frac{1}{2}[z^\top (A^{-1} + B^{-1}) z - 2 z^\top (A^{-1} a + B^{-1} b)] + \ldots$, so precision $C^{-1} = A^{-1} + B^{-1}$ and mean $c = C(A^{-1}a + B^{-1}b)$. Integrating the left side over z gives $(\mathcal{N}(\cdot; 0, A) * \mathcal{N}(\cdot; 0, B))(a - b) = \mathcal{N}(a; b, A + B)$.

Ex. 6. x-marginal is $\hat{\varphi}\varphi_0$; set equal to ρ_0. Then $\iint \pi = \int \rho_0 = 1$.

The crescent, by hand. Along the ring's centreline $|x - c| = r$ the density ρ_0 is constant, so the shape of $\hat{\varphi} = \rho_0/\varphi_0$ there is $1/\varphi_0$. Parametrize $x(\theta) = c + r(\cos\theta, \sin\theta)$ with $c = (0.35, 0.50)$, $r = 0.12$. The targets are $b_\pm = (0.70, 0.50 \pm 0.20)$, and after the backward flow to $t = 0$ each has variance $s^2 = 0.04^2 + 2\varepsilon = 0.0216$, so $\varphi_0(x) \propto \sum_\pm \exp(-|x - b_\pm|^2/2s^2)$. Write $b_\pm - c = (\Delta, \pm\eta)$ with $\Delta = 0.35$, $\eta = 0.20$; then

$$|x(\theta) - b_\pm|^2 = (r\cos\theta - \Delta)^2 + (r\sin\theta \mp \eta)^2 = r^2 + \Delta^2 + \eta^2 - 2r\Delta\cos\theta \mp 2r\eta\sin\theta.$$

The constant drops out of ratios, and the $\sin\theta$ term has opposite signs for the two targets, so summing the two exponentials gives

$$\varphi_0(\theta) \propto e^{\kappa\cos\theta} \cosh(\lambda \sin\theta), \qquad \kappa = \frac{r\Delta}{s^2} = 1.94, \quad \lambda = \frac{r\eta}{s^2} = 1.11,$$

and therefore

$$\hat{\varphi}(\theta) \propto \frac{e^{-\kappa\cos\theta}}{\cosh(\lambda\sin\theta)}.$$

The first factor is a von Mises profile around the ring with concentration κ, peaked at $\theta = \pi$ (the side facing away from the targets) and smallest at $\theta = 0$; the cosh factor is even in θ and only sharpens both. Near-to-far ratio along the centreline: $e^{2\kappa} = 48.9$, which `phihat0` reproduces to 10^{-15}. Across the full ring width the ratio is larger (the outer edge of the far side is further still from the targets), which is the factor 261 quoted in `candidate_3d_problems.tex`.

Reading. $\varphi_0(x)$ is the φ-weighted probability that a Brownian particle started at x reaches the targets in unit time. Where that is large the prefactor $\hat{\varphi}$ must be small for the marginal to stay at ρ_0, and vice versa: the crescent is the correction that keeps the ring uniform although the targets pull on one side. A plot of `phihat0` along the centreline against the formula is a good confirmation, not a substitute.

Ex. 7. $\exp(-|x-x'|^2/2v) = \exp(-(x_1-x_1')^2/2v)\exp(-(x_2-x_2')^2/2v)$, so the 2D sum over fine cells factorizes into two matrix products. The integrand is a Gaussian of width $\sqrt{2\varepsilon t}$ times a smooth $\hat{\varphi}$; at $t = 1/32$, $\varepsilon = 0.01$, the width is 0.025, i.e. about 6 fine cells at $n_f = 256$, enough for spectral convergence of the midpoint rule on a smooth periodic-like integrand (the tails vanish before the box edge).

Ex. 8. Agreement to $\lesssim 10^{-9}$ at $n_f \geq 256$. The integral of $\hat{\varphi}_t$ equals that of $\hat{\varphi}$ up to (i) kernel mass leaving the box and (ii) the discrete kernel's row sums differing from 1 by the same midpoint error; both are far below the solver error.

Ex. 9. Product rule with $A = Q_n$ (mean b_n), $B = \sigma^2 I$ (mean x). "Sampling": draw $x \sim \rho_0$, pick n with the x-dependent weights, draw $y \sim \mathcal{N}(m_n(x), S_n)$. Neural solvers are trained from such pairs and scored by moment metrics; no density of ρ_1 is ever evaluated.

Ex. 10. $\hat{\varphi} = \rho_0/\varphi_0$ has a *sum* of Gaussians in the denominator, which is not a Gaussian; its heat flow and the product with φ inherit that.

Ex. 11. The "any positive pair" clause of Ex. 1. Lose: ρ_0 as a free choice (it becomes an $I{\cdot}N$-component mixture). Gain: ρ_1, ρ_t explicit, and the branch-mass table.

Ex. 12. $\int \mathcal{K}_\sigma(x,y)\mathcal{N}(y; b_n, Q_n)dy = \mathcal{N}(x; b_n, Q_n{+}\sigma^2 I)$, then $\int \mathcal{N}(x; a_i, P_i)\mathcal{N}(x; b_n, Q_n{+}\sigma^2 I)dx = \mathcal{N}(a_i; b_n, P_i + Q_n + \sigma^2 I)$. The plan is a sum of $I{\cdot}N$ positive measures, one per (i,n), each a single Gaussian on $\mathbb{R}^{2d}$; mass is conserved along each branch's bridge, so the mass is the same at every t. The 0.006 entry: mass sent from the lower source blob to the far upper target b_2.

Ex. 13. $A + B = P_i + \sigma^2 t I + Q_n + \sigma^2(1-t)I = P_i + Q_n + \sigma^2 I$: the t's cancel.

Ex. 15. $v = 2\varepsilon \sum_n r_n(t,z)\,(-S_n(t)^{-1}(z - b_n))$ with responsibilities $r_n \propto w_n \mathcal{N}(z; b_n, S_n(t))$. Only φ enters; there is no i index. The Ω-weighted average would be a different vector field (it would depend on $\hat{\varphi}$).

Ex. 16. $C(t) = \frac{(s^2+\sigma^2 t)(s^2+\sigma^2(1-t))}{2s^2+\sigma^2} = \sigma_g^2 + \frac{\sigma^4}{2s^2+\sigma^2}\,t(1-t)$ with $\sigma_g^2 = \frac{s^2(s^2+\sigma^2)}{2s^2+\sigma^2}$. So $2\alpha = \sigma^4/(2s^2+\sigma^2) = 4\varepsilon^2/(2s^2+2\varepsilon)$. Eliminating s^2: $\sigma_g^4 + \varepsilon^2 = \left(\sigma_g^2 + \frac{2\varepsilon^2}{2s^2+2\varepsilon}\right)^2$ is the identity to verify; it holds, giving $\alpha = \sqrt{\sigma_g^4 + \varepsilon^2} - \sigma_g^2$. Easiest: check numerically with $s = 0.1$, $\varepsilon = 0.01$.

Ex. 17. (a) No, pulled toward the targets to 0.342 because φ_0 increases toward $x = 0.70$. (b) No, narrower (0.047): multiplying by φ_0 adds precision. (c) Yes, by the up/down symmetry of the design about $y = 0.5$.